overlapping as if they've just toppled upon each other in sequence. How did they all get so carefully set up in the first place?

A major bonus in certain cyclic and bouncing models is that they offer an opportunity to attribute that low initial entropy to something that happened before the bounce. The latest update to the ekpyrotic model, developed jointly by Paul Steinhardt and Anna Ijjas, explains the early universe's low entropy by effectively taking all the entropy from a tiny patch of the pre-bounce universe and setting that as the initial entropy of the entire observable universe today.

This new model (which is so new that it appeared on the scene during the writing of this book) has some significant advantages over previous versions of the ekpyrotic scenario. In particular, it doesn't require extra spatial dimensions or a singularity at the bounce. In fact, the contraction might be fairly mild—the reduction in size of the universe might be as low as a factor of two. The details are (obviously) complicated, but the basic idea is that what's really cycling is the mix of ingredients in the universe, and the way observers would perceive its evolution. As mentioned before, it's a scalar field filling the universe that drives the contraction/bounce, rather than a brane collision.

If this new cyclic model describes our universe, then in some far future epoch, we will start to see distant galaxies stop in their expansion and slowly turn around back toward us. It will look, at first, like the early stages of a Big Crunch, with the background radiation starting to heat up from "cool" to "this is not quite as cool" as the cosmos gets just a tiny bit more crowded. But just as we start to think maybe we should worry, we are, out of nowhere, suddenly and spectacularly obliterated when the scalar field violently converts its energy into radiation and starts the next Big Bang cycle of the universe.

Intriguingly, one aspect this brand-new hot-off-the-presses version of the ekpyrotic model shares with the old one is that rogue gravitational waves could be a kind of inter-universe signal. In the old version, it's conceivable some gravitational waves could pass through the bulk from another brane. In this one, since the cosmos never gets truly small during the bounce, gravitational waves might pass from one cycle to the next. These signals would be incredibly hard to find, but if they existed, they could present us with clues about a universe before our own.

Watch this space.